ghzfirodz

February 25, 2023

[507]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from google.colab import files

import os
```

[508]:
```
df = pd.read_csv("netflix.csv")
```

[509]:
```
df.columns
```

[509]:
```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')
```

[510]:
```
df.index
```

[510]:
```
RangeIndex(start=0, stop=8807, step=1)
```

[511]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
```

```
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

[512]:
```
df['date_added'] =  pd.to_datetime(df['date_added'])
```

[513]:
```
df.release_year.min()
```

[513]: 1925

[514]:
```
df.release_year.max()
```

[514]: 2021

[515]:
```
df.shape
```

[515]: (8807, 12)

[516]:
```
df.describe(include='all')
```

```
<ipython-input-516-174ba9bf1a5c>:1: FutureWarning: Treating datetime data as
categorical rather than numeric in `.describe` is deprecated and will be removed
in a future version of pandas. Specify `datetime_is_numeric=True` to silence
this warning and adopt the future behavior now.
  df.describe(include='all')
```

[516]:

	show_id	type	title	director	\
count	8807	8807	8807	6173	
unique	8807	2	8807	4528	
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	
freq	1	6131	1	19	
first	NaN	NaN	NaN	NaN	
last	NaN	NaN	NaN	NaN	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	cast	country	date_added	release_year	\
count	7982	7976	8797	8807.000000	
unique	7692	748	1714	NaN	
top	David Attenborough	United States	2020-01-01 00:00:00	NaN	
freq	19	2818	110	NaN	
first	NaN	NaN	2008-01-01 00:00:00	NaN	
last	NaN	NaN	2021-09-25 00:00:00	NaN	

```
mean                   NaN            NaN                   NaN   2014.180198
std                    NaN            NaN                   NaN      8.819312
min                    NaN            NaN                   NaN   1925.000000
25%                    NaN            NaN                   NaN   2013.000000
50%                    NaN            NaN                   NaN   2017.000000
75%                    NaN            NaN                   NaN   2019.000000
max                    NaN            NaN                   NaN   2021.000000

       rating  duration                     listed_in  \
count    8803      8804                          8807
unique     17       220                           514
top     TV-MA  1 Season  Dramas, International Movies
freq     3207      1793                           362
first     NaN       NaN                           NaN
last      NaN       NaN                           NaN
mean      NaN       NaN                           NaN
std       NaN       NaN                           NaN
min       NaN       NaN                           NaN
25%       NaN       NaN                           NaN
50%       NaN       NaN                           NaN
75%       NaN       NaN                           NaN
max       NaN       NaN                           NaN

                                                description
count                                                  8807
unique                                                 8775
top     Paranormal activity at a lush, abandoned prope…
freq                                                      4
first                                                   NaN
last                                                    NaN
mean                                                    NaN
std                                                     NaN
min                                                     NaN
25%                                                     NaN
50%                                                     NaN
75%                                                     NaN
max                                                     NaN
```

[517]:
```
df['country']=df['country'].astype('category')
df['listed_in']=df['listed_in'].astype('category')
```

[518]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
```

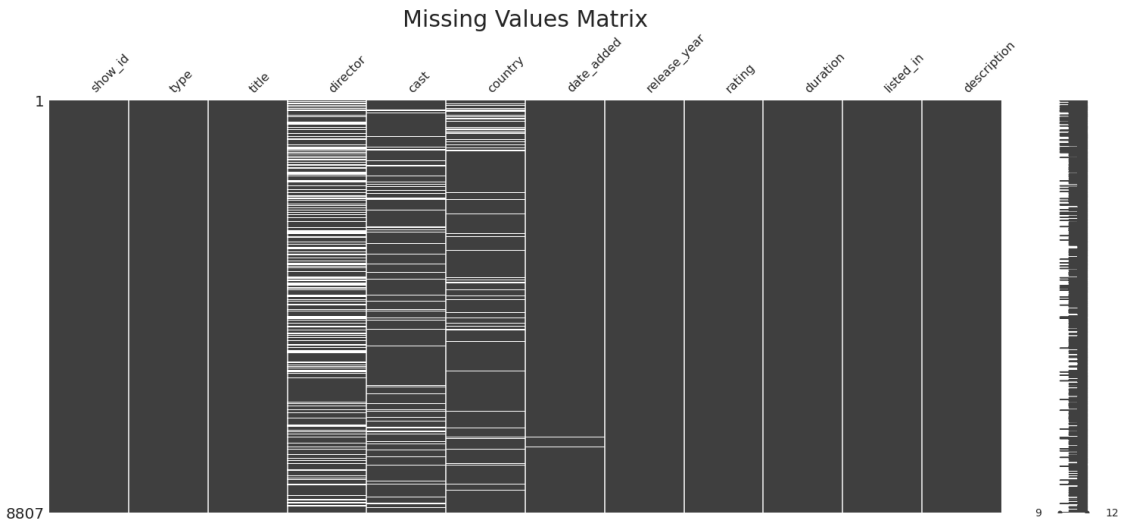
```
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   category
 6   date_added    8797 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   category
 11  description   8807 non-null   object
dtypes: category(2), datetime64[ns](1), int64(1), object(8)
memory usage: 764.8+ KB
```

[519]:
```
for i in df.columns:
    null_rate = df[i].isna().sum() / len(df) * 100
    if null_rate >= 0 :
        print("{} null rate: {}%".format(i,round(null_rate,2)))
```

```
show_id null rate: 0.0%
type null rate: 0.0%
title null rate: 0.0%
director null rate: 29.91%
cast null rate: 9.37%
country null rate: 9.44%
date_added null rate: 0.11%
release_year null rate: 0.0%
rating null rate: 0.05%
duration null rate: 0.03%
listed_in null rate: 0.0%
description null rate: 0.0%
```

[520]:
```
import missingno as msno
msno.matrix(df)
plt.title("Missing Values Matrix",fontsize=30)
```

[520]:
```
Text(0.5, 1.0, 'Missing Values Matrix')
```



[521]:
```
df.director.value_counts()
```

[521]:
```
Rajiv Chilaka                     19
Raúl Campos, Jan Suter            18
Marcus Raboy                      16
Suhas Kadav                       16
Jay Karas                         14
                                  ..
Raymie Muzquiz, Stu Livingston     1
Joe Menendez                       1
Eric Bross                         1
Will Eisenberg                     1
Mozez Singh                        1
Name: director, Length: 4528, dtype: int64
```

[522]:
```
df.director.nunique()
```

[522]:
```
4528
```

[523]:
```
df['cast']=df['cast'].str.split(',')
df['listed_in']=df['listed_in'].str.split(',')
df['country']=df['country'].str.split(',')
df['director']=df['director'].str.split(',')
```

[524]:
```
df = df.explode("cast")
df = df.explode("country")
df = df.explode("director")
df = df.explode("listed_in").reset_index(drop=True)
df['cast'] = df['cast'].str.strip()
df['listed_in'] = df['listed_in'].str.strip()
```

```
df['director'] = df['director'].str.strip()
df['country'] = df['country'].str.strip()
```

[525]:
```
df.shape
```

[525]: (202065, 12)

BEFORE DATA CLEANING

[526]:
```
for i in df.columns:
    null_rate = df[i].isna().sum() / len(df) * 100
    if null_rate >= 0 :
        print("{} null rate: {}%".format(i,round(null_rate,2)))
```

```
show_id null rate: 0.0%
type null rate: 0.0%
title null rate: 0.0%
director null rate: 25.06%
cast null rate: 1.06%
country null rate: 5.89%
date_added null rate: 0.08%
release_year null rate: 0.0%
rating null rate: 0.03%
duration null rate: 0.0%
listed_in null rate: 0.0%
description null rate: 0.0%
```

[527]:
```
df['date_added'] = df.groupby(['release_year','type'])['date_added'].
↪transform(lambda x: x.fillna(x.mode().iloc[0]))
```

[528]:
```
df['country'] = df.groupby(['listed_in'])['country'].transform(lambda x: x.
↪fillna(x.mode().iloc[0]))
```

[529]:
```
df['director'] = df.groupby(['release_year'])['director'].transform(lambda x: x.
↪fillna(x.mode()[0]) if len(x.mode()) > 0 else x)
```

[530]:
```
df['cast'] = df.groupby(['release_year'])['cast'].transform(lambda x: x.
↪fillna(x.mode()[0]) if len(x.mode()) > 0 else x)
```

[531]:
```
df = df.dropna(subset=['duration',])
```

[532]:
```
df.drop(df[(df['release_year'] == 1925) & df.isnull().any(axis=1)].index,␣
↪inplace=True)
```

```
/usr/local/lib/python3.8/dist-packages/pandas/core/frame.py:4906:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  return super().drop(
```

[533]:
```
df['day_of_week'] = df['date_added'].dt.day_name()
```

[534]:
```
df['rating'].fillna("Not Rated",inplace=True)
df["year_added"] = df['date_added'].dt.year
df["year_added"] = df["year_added"].astype("Int64")
df["month_added"] = df['date_added'].dt.month
df["month_added"] = df["month_added"].astype("Int64")
```

AFTER DATA CLEANING"

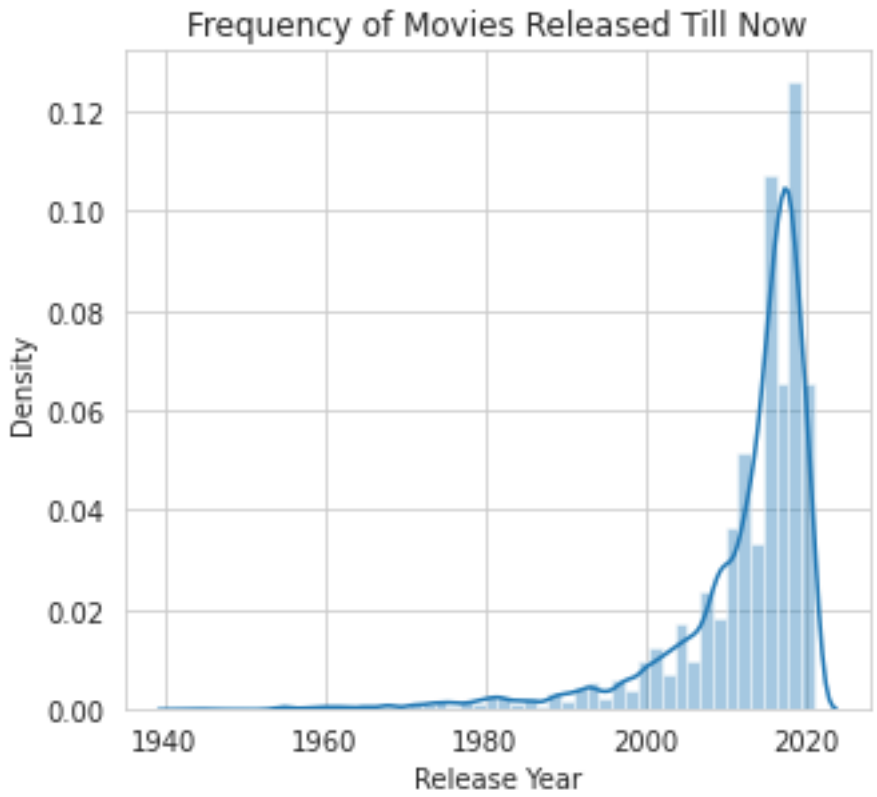
[535]:
```
for i in df.columns:
    null_rate = df[i].isna().sum() / len(df) * 100
    if null_rate >= 0 :
        print("{} null rate: {}%".format(i,round(null_rate,2)))
```

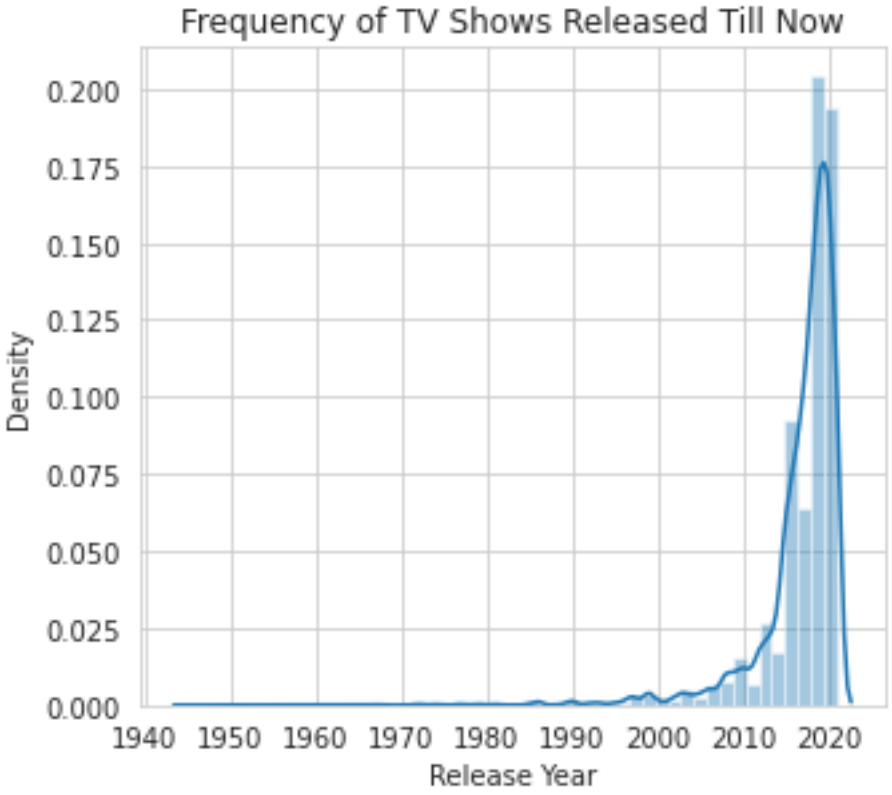
```
show_id null rate: 0.0%
type null rate: 0.0%
title null rate: 0.0%
director null rate: 0.0%
cast null rate: 0.0%
country null rate: 0.0%
date_added null rate: 0.0%
release_year null rate: 0.0%
rating null rate: 0.0%
duration null rate: 0.0%
listed_in null rate: 0.0%
description null rate: 0.0%
day_of_week null rate: 0.0%
year_added null rate: 0.0%
month_added null rate: 0.0%
```

[536]:
```
movies = df[df['type'] == 'Movie']
tv_shows = df[df['type'] == 'TV Show']

# create two subplots for movies and tv shows
fig, axes = plt.subplots(nrows=2, figsize=(5, 10))
sns.distplot(movies['release_year'], ax=axes[0])
sns.distplot(tv_shows['release_year'], ax=axes[1])

# set titles and labels for the subplots
axes[0].set_title('Frequency of Movies Released Till Now')
axes[1].set_title('Frequency of TV Shows Released Till Now')
axes[0].set_xlabel('Release Year')
axes[1].set_xlabel('Release Year')
```

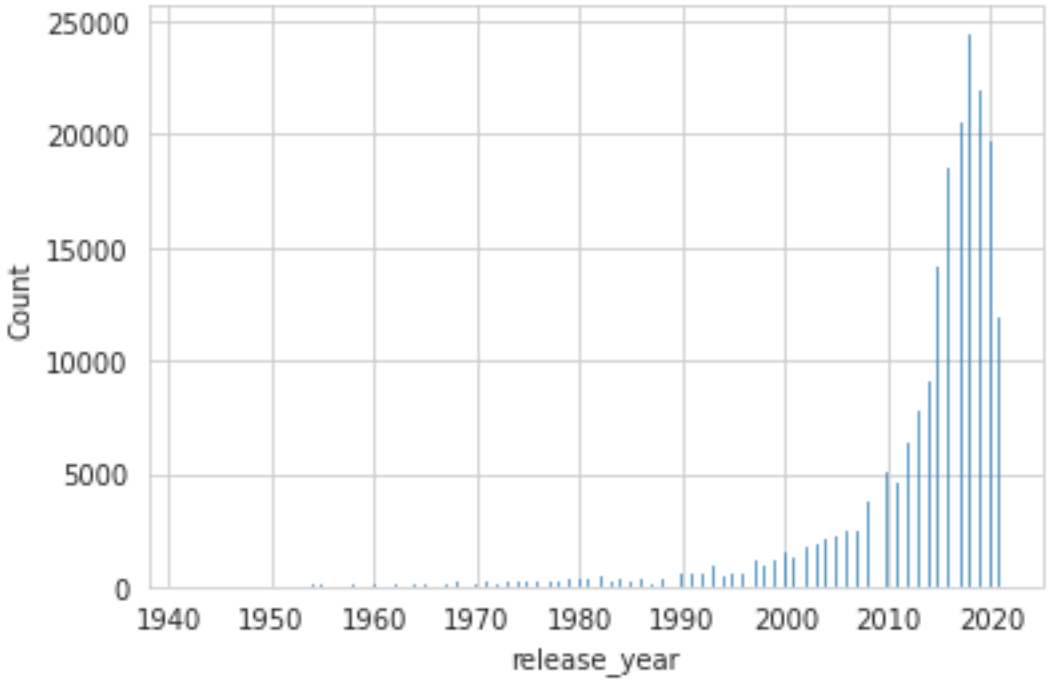
```
axes[0].set_ylabel('Density')
axes[1].set_ylabel('Density')

plt.show()
```

```
/usr/local/lib/python3.8/dist-packages/seaborn/distributions.py:2619:
FutureWarning: `distplot` is a deprecated function and will be removed in a
future version. Please adapt your code to use either `displot` (a figure-level
function with similar flexibility) or `histplot` (an axes-level function for
histograms).
  warnings.warn(msg, FutureWarning)
/usr/local/lib/python3.8/dist-packages/seaborn/distributions.py:2619:
FutureWarning: `distplot` is a deprecated function and will be removed in a
future version. Please adapt your code to use either `displot` (a figure-level
function with similar flexibility) or `histplot` (an axes-level function for
histograms).
  warnings.warn(msg, FutureWarning)
```


Frequency of Movies Released Till Now
Density
0.12
0.10
0.08
0.06
0.04
0.02
0.00
1940
1960
1980
2000
2020
Release Year


Frequency of TV Shows Released Till Now
Density
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1940
1950
1960
1970
1980
1990
2000
2010
2020
Release Year

[537]:
```
ax = sns.histplot(data=df, x="release_year")
plt.show()
```

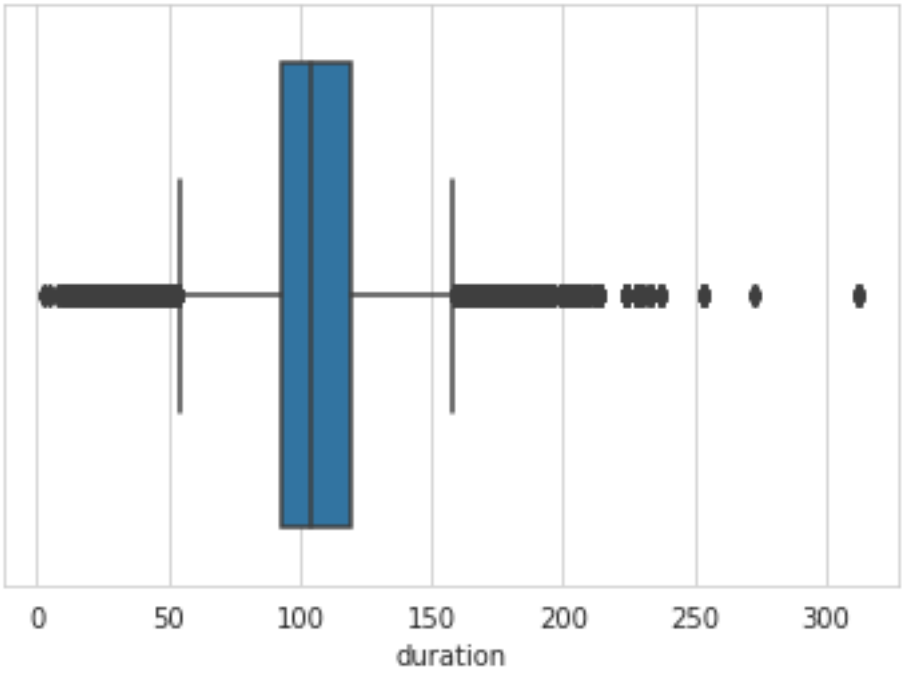


[538]:
```
df1 = df.loc[df['type'] =='Movie']
df1['duration'] = df1['duration'].str.replace(" min","").astype("float").
  ↪astype("int")
df1['duration'].value_counts()
sns.boxplot(data=df1, x="duration")
df1['duration'].max()
df1['duration'].describe()
```

```
<ipython-input-538-4ed8347f45a5>:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df1['duration'] = df1['duration'].str.replace("
min","").astype("float").astype("int")
```

[538]:
```
count    145914.000000
mean        106.840385
std          24.709395
min           3.000000
25%          93.000000
```

```
50%          104.000000
75%          119.000000
max          312.000000
Name: duration, dtype: float64
```



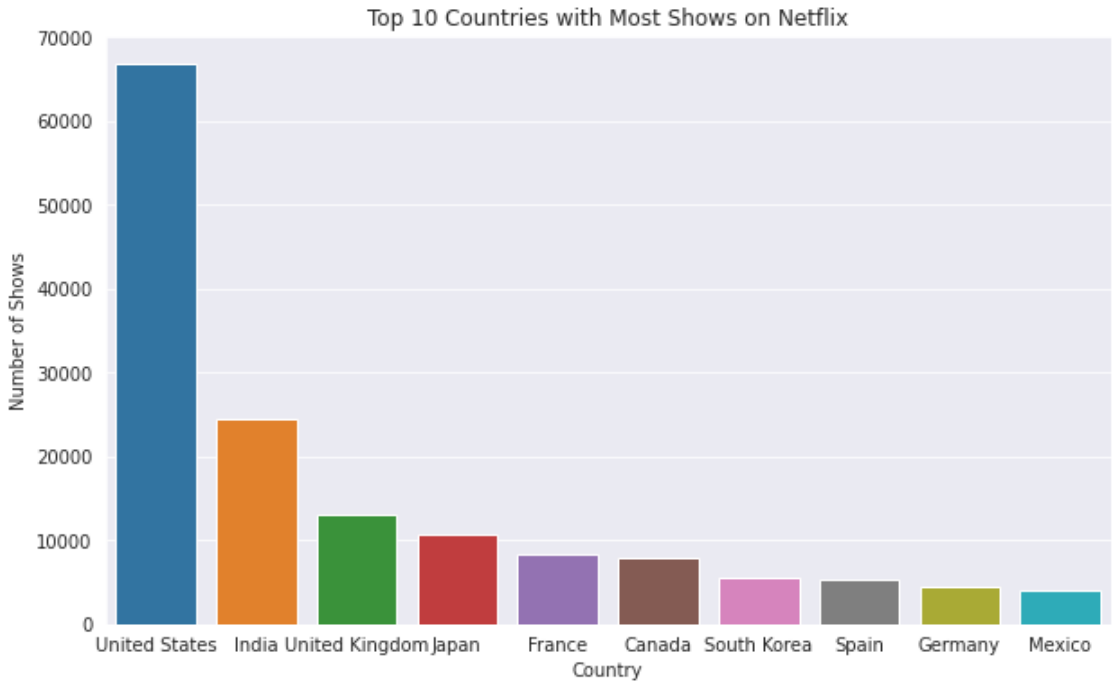
[539]:
```
df1.loc[df1['duration']==df1['duration'].max()].iloc[0]['title']
```

[539]:
```
'Black Mirror: Bandersnatch'
```

[540]:
```
df1.loc[df1['duration']==df1['duration'].min()].iloc[0]['title']
```

[540]:
```
'Silent'
```

[541]:
```
country_grouped = df.groupby('country').count().sort_values(by='show_id',␣
↪ascending=False)[:10]
country_grouped
sns.set_style('darkgrid')
plt.figure(figsize=(10, 6))
sns.barplot(x=country_grouped.index, y='show_id', data=country_grouped)
plt.title('Top 10 Countries with Most Shows on Netflix')
plt.xlabel('Country')
plt.ylabel('Number of Shows')
plt.show()
country_grouped.index
```



[541]:
```
Index(['United States', 'India', 'United Kingdom', 'Japan', 'France', 'Canada',
       'South Korea', 'Spain', 'Germany', 'Mexico'],
      dtype='object', name='country')
```

#How has the number of movies released per year changed over the last 20-30 years?

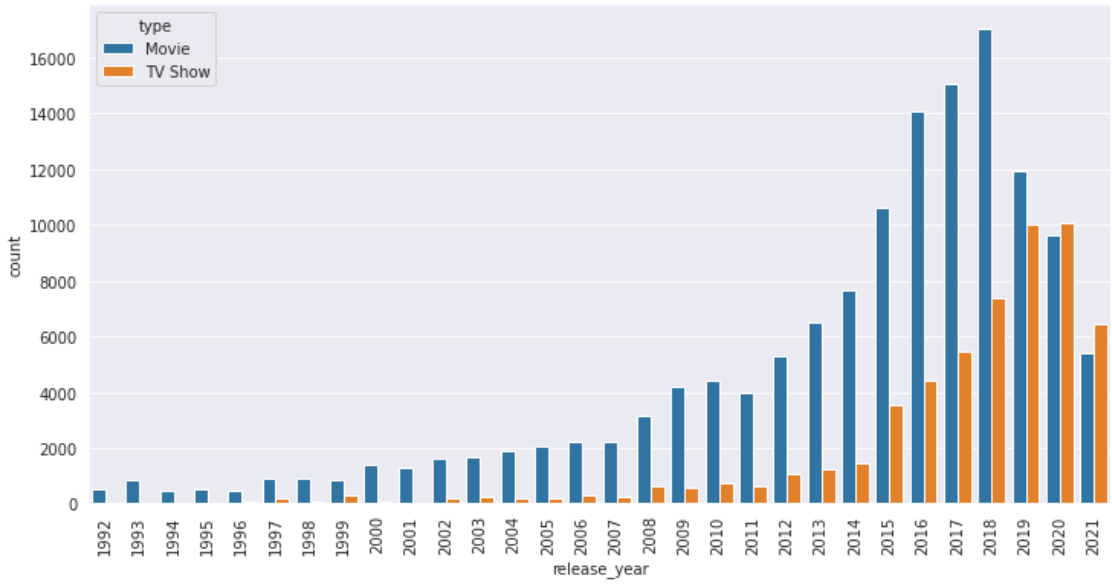
[542]:
```
df1 = df.loc[df['release_year'] > df['release_year'].max() - 30,␣
 ↪['release_year', 'type']]
```

[543]:
```
df1['type'].value_counts()
```

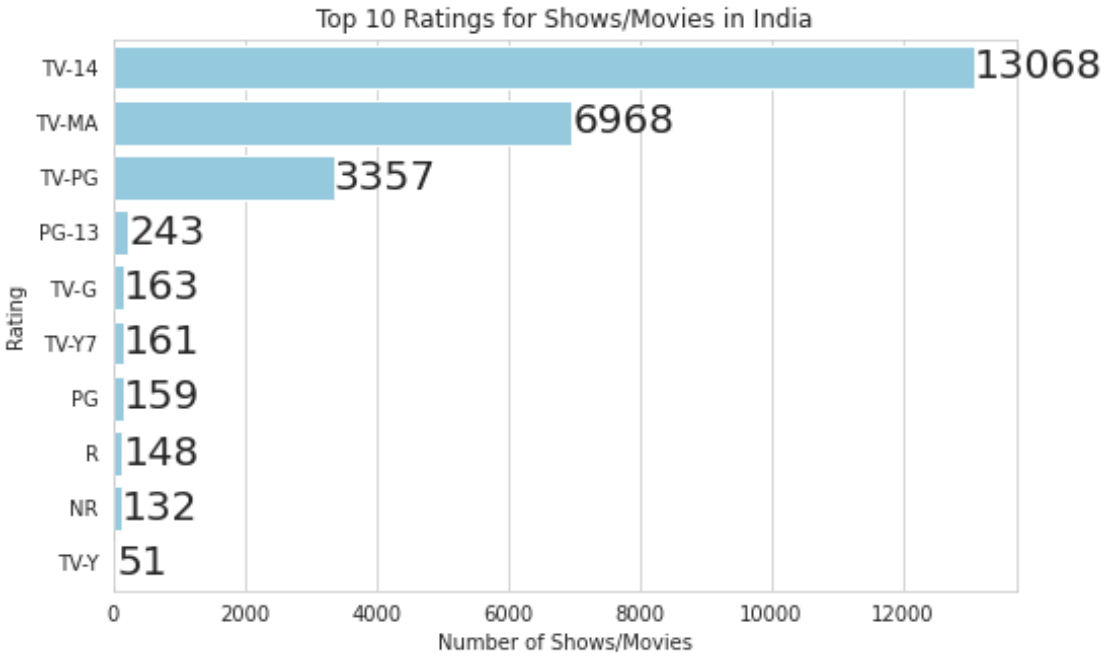
[543]:
```
Movie      138820
TV Show     55797
Name: type, dtype: int64
```

[544]:
```
# create the count plot
plt.figure(figsize=(12, 6))
plt.xticks(rotation=90)
sns.countplot(x='release_year', hue='type', data=df1)
```

[544]:
```
<AxesSubplot:xlabel='release_year', ylabel='count'>
```



[545]:
```
rating_groupby = df.groupby(['rating','country'])['title'].count()
india_ratings = rating_groupby.loc[(slice(None), 'India')].
 ↪sort_values(ascending=False)
india_ratings = india_ratings.iloc[:10]
india_ratings = india_ratings.rename_axis(['rating']).reset_index()
india_ratings

sns.set_style('whitegrid')
sns.set_color_codes('pastel')

fig, ax = plt.subplots(figsize=(8, 5))
sns.barplot(x='title', y='rating', data=india_ratings, color='skyblue', ax=ax)

ax.set_title('Top 10 Ratings for Shows/Movies in India')
ax.set_xlabel('Number of Shows/Movies')
ax.set_ylabel('Rating')

# Add count as text
for i, v in enumerate(india_ratings['title']):
    ax.annotate(str(v), xy=(v + 1, i), va='center', fontsize=20)

plt.show()
```



[546]:
```
rating_groupby = df.groupby(['rating','country'])['title'].count()
general_audience = rating_groupby.loc['G', (slice(None))].
 ↪sort_values(ascending=False)
general_audience = general_audience.iloc[:10]
general_audience
```

[546]:
```
rating  country
G       United States     907
        United Kingdom    130
        Spain              74
        Ireland            65
        Germany            56
        Canada             48
        France             36
        East Germany       30
        Japan              30
        West Germany       30
Name: title, dtype: int64
```

[547]:
```
genre_groupby = df.groupby(['listed_in','country'])['title'].count()
general_audience = genre_groupby.loc['International Movies'].
 ↪sort_values(ascending=False)
general_audience = general_audience.iloc[:10]
general_audience
```

[547]:
```
country
India             8657
France            1740
United States     1211
United Kingdom    1199
Spain             1131
Japan              810
Egypt              787
Nigeria            784
Indonesia          727
Turkey             712
Name: title, dtype: int64
```

[548]:
```
type_groupby = df.groupby(['type','country'])['title'].count()
type_groupby = type_groupby.loc['TV Show'].sort_values(ascending=False)
type_groupby
```

[548]:
```
country
United States     16478
Japan              7039
United Kingdom     4417
South Korea        4344
Mexico             2221
                  …
Cyprus               10
                      8
Greece                6
Belarus               6
Uruguay               3
Name: title, Length: 66, dtype: int64
```

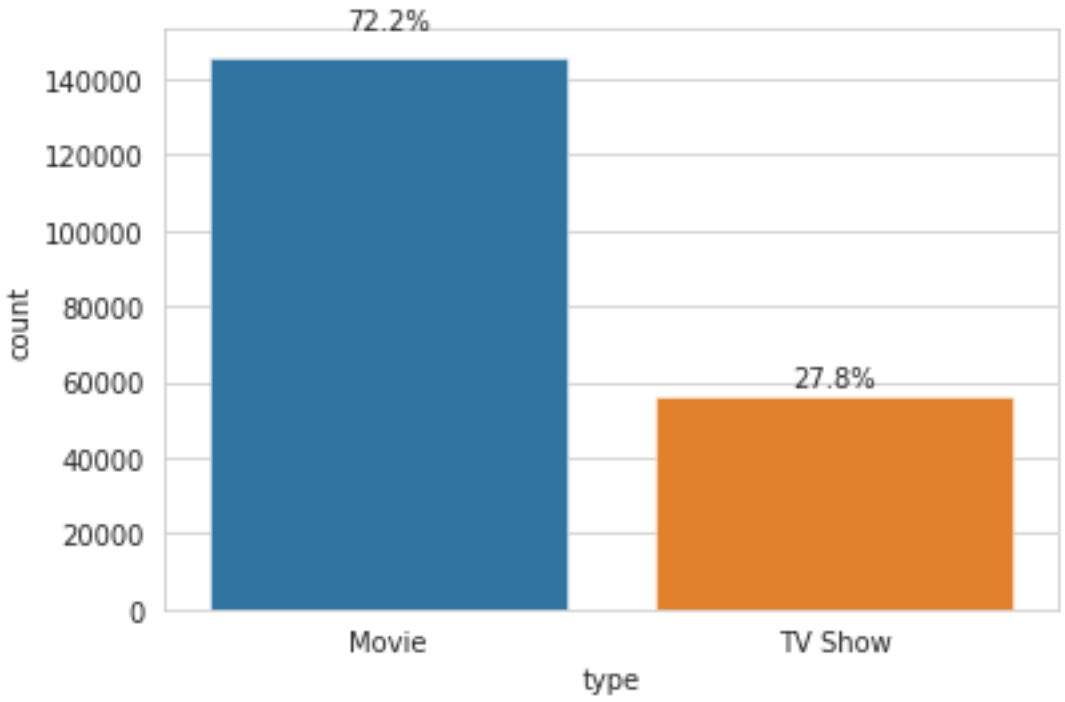
Explanation:- For the last 30 The Gradual Rate of movies has always been increasing but there was a dip from the year 2018

#Comparison of tv shows vs. movies

[549]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# create countplot
ax = sns.countplot(x='type', data=df, linewidth=0.5)

# get counts and calculate percentages
counts = df['type'].value_counts()
total = counts.sum()
percentages = counts / total * 100
print("percentages",percentages)
```

```
# add percentage labels
for i, p in enumerate(ax.patches):
    height = p.get_height()
    width = p.get_width()
    x, y = p.get_xy()
    ax.text(x + width/2 , y + height * 1.05, f'{percentages[i]:.1f}%',␣
 ↪ha='center')

plt.show()
```

```
percentages Movie      72.212847
TV Show    27.787153
Name: type, dtype: float64
```



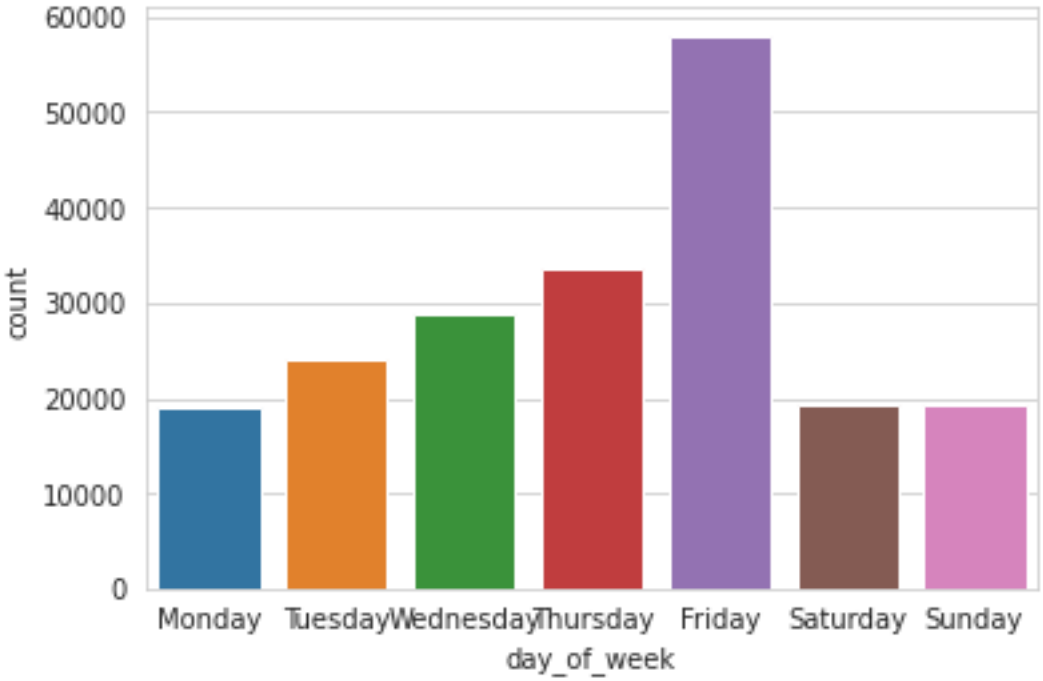
Netflix has more number of Movies than **TV**

#What was the best time to launch a TV show?

[550]:
```
df['day_of_week'] = df['date_added'].dt.day_name()
df1 = df.loc[df['type']=='TV Show']
```

[551]:
```
sns.countplot(x=df["day_of_week"], order=["Monday", "Tuesday", "Wednesday",␣
 ↪"Thursday", "Friday", "Saturday", "Sunday"])
```

[551]: <AxesSubplot:xlabel='day_of_week', ylabel='count'>



The Best Time To Launch A TV show would be Friday**

[552]:
```
data_corr = df.corr()
data_corr
```

[552]:

	release_year	year_added	month_added
release_year	1.000000	0.053252	-0.032102
year_added	0.053252	1.000000	-0.167706
month_added	-0.032102	-0.167706	1.000000

[553]:
```
sns.heatmap(data = df.corr())
```

[553]: <AxesSubplot:>

1 Analysis of Actors and Directors

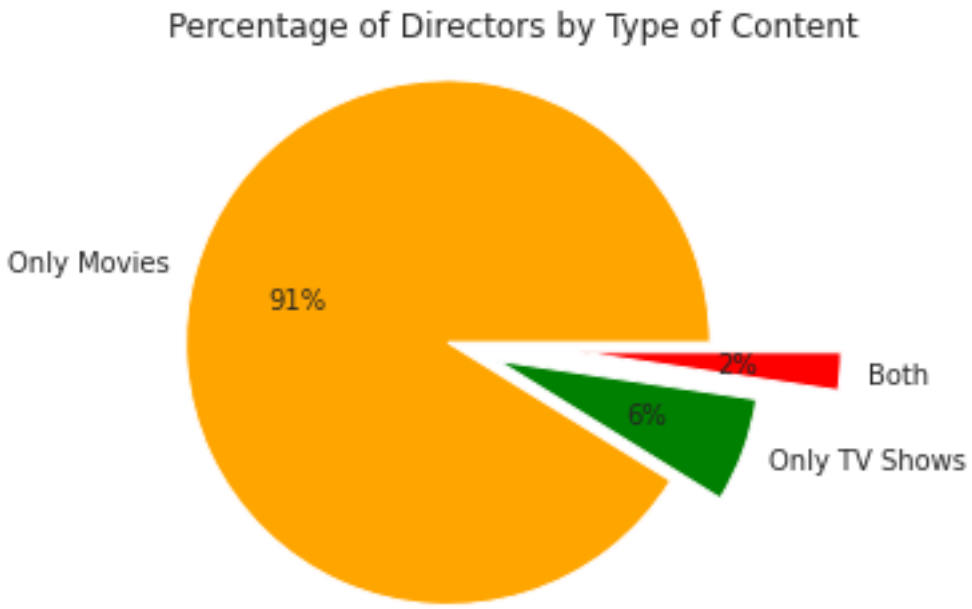
[554]:
```
df['duration'].value_counts()
```

[554]:
```
1 Season      35034
2 Seasons      9559
3 Seasons      5084
94 min         4343
106 min        4040
              ...
3 min             4
5 min             3
11 min            2
8 min             2
9 min             2
Name: duration, Length: 220, dtype: int64
```

[555]:
```
# Count the number of directors in each category
movie_directors = df[df['type'] == 'Movie']['director'].nunique()
tv_directors = df[df['type'] == 'TV Show']['director'].nunique()
both_directors = df.groupby('director')['type'].nunique().eq(2).sum()


# Calculate the percentages
```

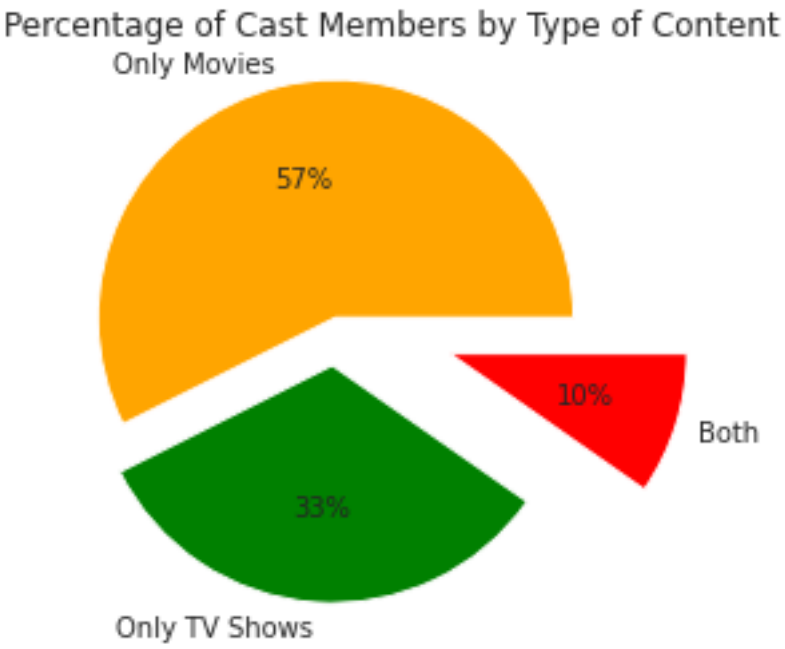
```
total_directors = df['director'].nunique()
movie_directors_percent = 100 * movie_directors / total_directors
tv_directors_percent = 100 * tv_directors / total_directors
both_directors_percent = 100 * both_directors / total_directors

# Print the results
print(f"Percentage of directors who directed only movies:␣
 ↪{movie_directors_percent:.2f}%")
print(f"Percentage of directors who directed only TV shows:␣
 ↪{tv_directors_percent:.2f}%")
print(f"Percentage of directors who directed both movies and TV shows:␣
 ↪{both_directors_percent:.2f}%")
explode = [0, 0.2, 0.5]
labels = ['Only Movies', 'Only TV Shows', 'Both']
sizes = [movie_directors_percent, tv_directors_percent, both_directors_percent]
colors = ['orange','green','red']
plt.pie(sizes, labels=labels, colors=colors,explode=explode, autopct='%.0f%%')
plt.axis('equal')
plt.title('Percentage of Directors by Type of Content')
plt.show()
```

```
Percentage of directors who directed only movies: 95.67%
Percentage of directors who directed only TV shows: 6.79%
Percentage of directors who directed both movies and TV shows: 2.46%
```



[556]:
```
# Count the number of cast members in each category
movie_cast = df[df['type'] == 'Movie']['cast'].nunique()
tv_cast = df[df['type'] == 'TV Show']['cast'].nunique()
both_cast = df.groupby('cast')['type'].nunique().eq(2).sum()

# Calculate the percentages
total_cast = df['cast'].nunique()
movie_cast_percent = 100 * movie_cast / total_cast
tv_cast_percent = 100 * tv_cast / total_cast
both_cast_percent = 100 * both_cast / total_cast

# Print the results
print(f"Percentage of cast members who acted only in movies:␣
 ↪{movie_cast_percent:.2f}%")
print(f"Percentage of cast members who acted only in TV shows: {tv_cast_percent:
 ↪.2f}%")
print(f"Percentage of cast members who acted in both movies and TV shows:␣
 ↪{both_cast_percent:.2f}%")

explode = [0, 0.2, 0.5]
labels = ['Only Movies', 'Only TV Shows', 'Both']
sizes = [movie_cast_percent, tv_cast_percent, both_cast_percent]
colors = ['orange','green','red']
plt.pie(sizes, labels=labels, colors=colors,explode=explode, autopct='%.0f%%')
plt.axis('equal')
plt.title('Percentage of Cast Members by Type of Content')
plt.show()
```

```
Percentage of cast members who acted only in movies: 71.22%
Percentage of cast members who acted only in TV shows: 40.82%
Percentage of cast members who acted in both movies and TV shows: 12.04%
```



2 Does Netflix has more focus on TV Shows than movies in recent years

[557]:
```
df1 = df.loc[df['release_year'] > df['release_year'].max() - 10,␣
↪['release_year', 'type']]
df1
```

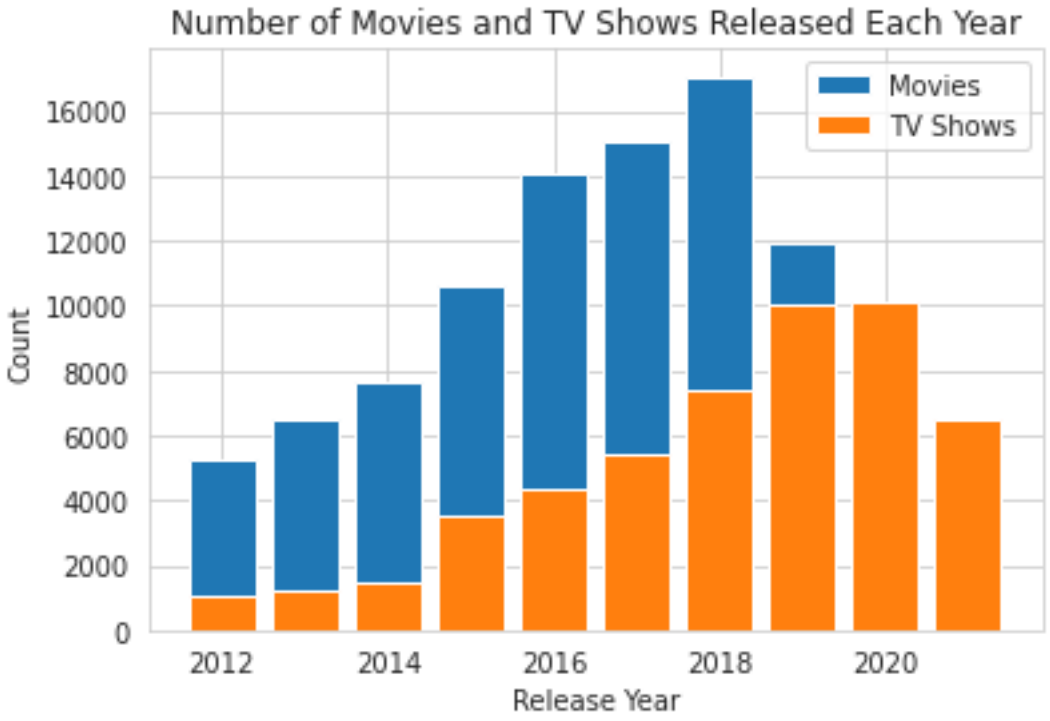
[557]:

	release_year	type
0	2020	Movie
1	2021	TV Show
2	2021	TV Show
3	2021	TV Show
4	2021	TV Show
…	…	…
202060	2015	Movie
202061	2015	Movie
202062	2015	Movie
202063	2015	Movie
202064	2015	Movie

[154267 rows x 2 columns]

[558]:
```
import matplotlib.pyplot as plt
```

```
grouped = df1.groupby(['release_year', 'type']).size().reset_index(name='count')

# Create separate DataFrames for movies and TV shows
movies = grouped[grouped['type'] == 'Movie']
tv_shows = grouped[grouped['type'] == 'TV Show']

# Plot the count of movies and TV shows over time
plt.bar(movies['release_year'], movies['count'], label='Movies')
plt.bar(tv_shows['release_year'], tv_shows['count'], label='TV Shows')

plt.xlabel('Release Year')
plt.ylabel('Count')
plt.title('Number of Movies and TV Shows Released Each Year')
plt.legend()
plt.show()
```



**YES NETFLIX HAS BEEN FOCUSING MORE ON TV SHOWS THAN MOVIES THE THE GROWTH RATE HAS ALWAYS BEEN IN A POSITIVE WAY FOR TV SHOWS THE DOWN-FALL OF MOVIES STARTED IN 2018 which shows more audience are connected to TV SHOWS THAN MOVIES

Understanding what content is available in different countries

[559]:
```
df.head()
```

[559]:
```
  show_id     type                  title             director         cast  \
0      s1    Movie  Dick Johnson Is Dead      Kirsten Johnson   Adil Dehbi
1      s2  TV Show         Blood & Water  Rathindran R Prasad   Ama Qamata
2      s2  TV Show         Blood & Water  Rathindran R Prasad   Ama Qamata
3      s2  TV Show         Blood & Water  Rathindran R Prasad   Ama Qamata
4      s2  TV Show         Blood & Water  Rathindran R Prasad  Khosi Ngema

         country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
2   South Africa 2021-09-24          2021  TV-MA  2 Seasons
3   South Africa 2021-09-24          2021  TV-MA  2 Seasons
4   South Africa 2021-09-24          2021  TV-MA  2 Seasons

                listed_in                                        description  \
0           Documentaries  As her father nears the end of his life, filmm…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
2              TV Dramas  After crossing paths at a party, a Cape Town t…
3            TV Mysteries  After crossing paths at a party, a Cape Town t…
4  International TV Shows  After crossing paths at a party, a Cape Town t…

  day_of_week  year_added  month_added
0    Saturday        2021            9
1      Friday        2021            9
2      Friday        2021            9
3      Friday        2021            9
4      Friday        2021            9
```

[560]:
```
import pandas as pd

# assuming your DataFrame is stored in a variable called 'df'
pivoted_df = pd.pivot_table(df, index='country', columns='listed_in',␣
 ↪values='title', aggfunc='count')
pivoted_df = pivoted_df.fillna(0)
pivoted_df['Total_Movies'] = pivoted_df.sum(axis=1)

pivoted_df = pivoted_df.sort_values(by='Total_Movies', ascending=False)
top_10_countries = pivoted_df.head(10)
bottom_10_countries = pivoted_df.tail(10)

top_10_countries = top_10_countries.drop('Total_Movies', axis=1)
bottom_10_countries = bottom_10_countries.drop('Total_Movies', axis=1)


bottom_10_countries
```

[560]:
listed_in	Action & Adventure	Anime Features	Anime Series
country			
Armenia	0.0	0.0	0.0
Palestine	0.0	0.0	0.0
Panama	0.0	0.0	0.0
Afghanistan	0.0	0.0	0.0
Mongolia	0.0	0.0	0.0
Botswana	0.0	0.0	0.0
Samoa	0.0	0.0	0.0
Uganda	0.0	0.0	0.0
Kazakhstan	0.0	0.0	0.0
Nicaragua	0.0	0.0	0.0

listed_in	British TV Shows	Children & Family Movies	Classic & Cult TV
country			
Armenia	0.0	0.0	0.0
Palestine	0.0	0.0	0.0
Panama	0.0	0.0	0.0
Afghanistan	0.0	0.0	0.0
Mongolia	0.0	0.0	0.0
Botswana	0.0	0.0	0.0
Samoa	0.0	0.0	0.0
Uganda	0.0	0.0	0.0
Kazakhstan	0.0	1.0	0.0
Nicaragua	0.0	0.0	0.0

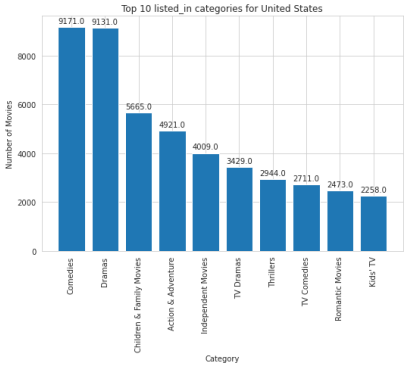
listed_in	Classic Movies	Comedies	Crime TV Shows	Cult Movies	…
country					…
Armenia	0.0	0.0	0.0	0.0	…
Palestine	0.0	0.0	0.0	0.0	…
Panama	0.0	0.0	0.0	0.0	…
Afghanistan	0.0	0.0	0.0	0.0	…
Mongolia	0.0	0.0	0.0	0.0	…
Botswana	0.0	0.0	0.0	0.0	…
Samoa	0.0	0.0	0.0	0.0	…
Uganda	0.0	0.0	0.0	0.0	…
Kazakhstan	0.0	0.0	0.0	0.0	…
Nicaragua	0.0	0.0	0.0	0.0	…

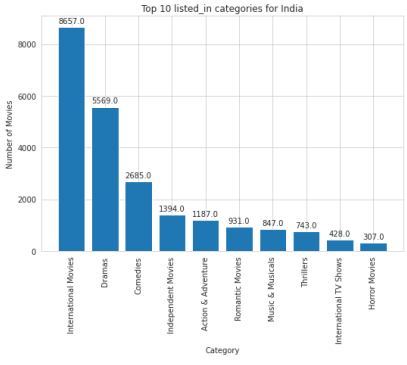
listed_in	TV Action & Adventure	TV Comedies	TV Dramas	TV Horror
country				
Armenia	0.0	0.0	0.0	0.0
Palestine	0.0	0.0	0.0	0.0
Panama	0.0	0.0	0.0	0.0
Afghanistan	0.0	0.0	0.0	0.0
Mongolia	0.0	0.0	0.0	0.0
Botswana	0.0	0.0	0.0	0.0

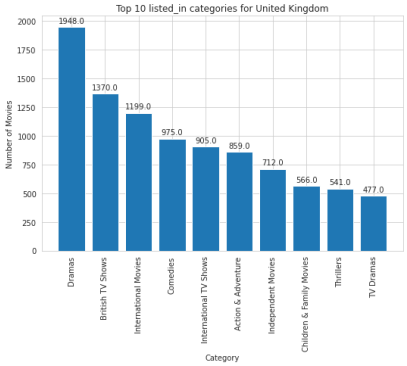
```
Samoa                          0.0          0.0        0.0        0.0
Uganda                         0.0          0.0        0.0        0.0
Kazakhstan                     0.0          0.0        0.0        0.0
Nicaragua                      0.0          0.0        0.0        0.0

listed_in    TV Mysteries  TV Sci-Fi & Fantasy  TV Shows  TV Thrillers  \
country
Armenia               0.0                  0.0       0.0           0.0
Palestine             0.0                  0.0       0.0           0.0
Panama                0.0                  0.0       0.0           0.0
Afghanistan           0.0                  0.0       0.0           0.0
Mongolia              0.0                  0.0       0.0           0.0
Botswana              0.0                  0.0       0.0           0.0
Samoa                 0.0                  0.0       0.0           0.0
Uganda                0.0                  0.0       0.0           0.0
Kazakhstan            0.0                  0.0       0.0           0.0
Nicaragua             0.0                  0.0       0.0           0.0

listed_in    Teen TV Shows  Thrillers
country
Armenia                0.0        0.0
Palestine              0.0        0.0
Panama                 0.0        0.0
Afghanistan            0.0        0.0
Mongolia               0.0        0.0
Botswana               0.0        0.0
Samoa                  0.0        0.0
Uganda                 0.0        0.0
Kazakhstan             0.0        0.0
Nicaragua              0.0        0.0

[10 rows x 42 columns]
```

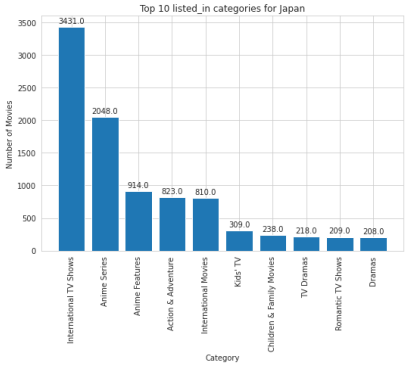
[561]:
```python
import matplotlib.pyplot as plt

# Drop the 'Total_Movies' column from the DataFrame

fig, axes = plt.subplots(nrows=5, ncols=2, figsize=(15, 35))
for i, country in enumerate(top_10_countries.index):
    row = i // 2
    col = i % 2
    # select the top 10 listed_in categories for the current country
    top_10_categories = top_10_countries.loc[country].
 ↪sort_values(ascending=False)[:10]
    # create a bar chart for the current country
    bars = axes[row, col].bar(top_10_categories.index, top_10_categories.values)
    axes[row, col].set_title(f'Top 10 listed_in categories for {country}')
```

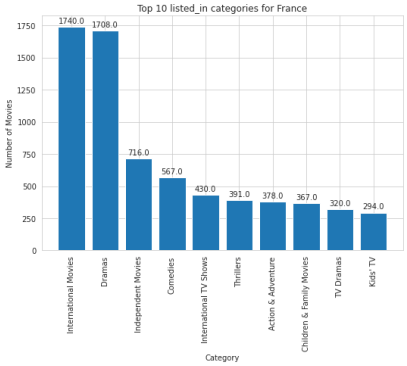
```
    axes[row, col].set_xlabel('Category')
    axes[row, col].set_ylabel('Number of Movies')
    axes[row, col].tick_params(axis='x', rotation=90)
    # add count labels to each bar
    for bar in bars:
        height = bar.get_height()
        axes[row, col].annotate(f'{height}', xy=(bar.get_x() + bar.get_width() /
 ↪ 2, height), xytext=(0, 3),
                                textcoords="offset points", ha='center',␣
 ↪va='bottom')
plt.tight_layout()
plt.show()
```

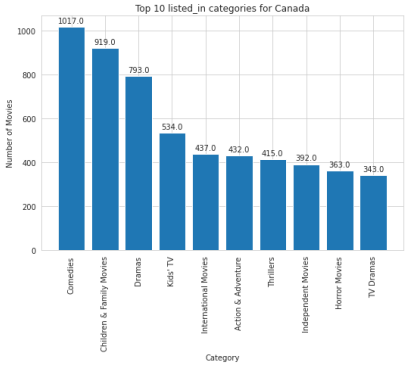

Top 10 listed_in categories for United States
Number of Movies
Category
Comedies
Dramas
Children & Family Movies
Action & Adventure
Independent Movies
TV Dramas
Thrillers
TV Comedies
Romantic Movies
Kids' TV
9171.0
9131.0
5665.0
4921.0
4009.0
3429.0
2944.0
2711.0
2473.0
2258.0

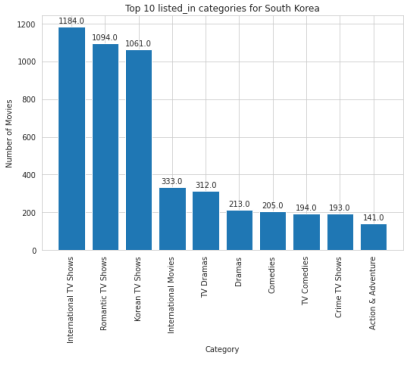

Top 10 listed_in categories for India
Number of Movies
Category
International Movies
Dramas
Comedies
Independent Movies
Action & Adventure
Romantic Movies
Music & Musicals
Thrillers
International TV Shows
Horror Movies
8657.0
5569.0
2685.0
1394.0
1187.0
931.0
847.0
743.0
428.0
307.0

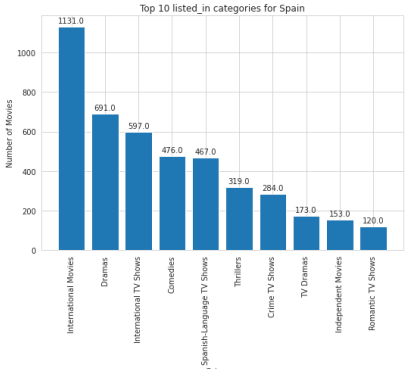

Top 10 listed_in categories for United Kingdom
Number of Movies
Category
Dramas
British TV Shows
International Movies
Comedies
International TV Shows
Action & Adventure
Independent Movies
Children & Family Movies
Thrillers
TV Dramas
1948.0
1370.0
1199.0
975.0
905.0
859.0
712.0
566.0
541.0
477.0

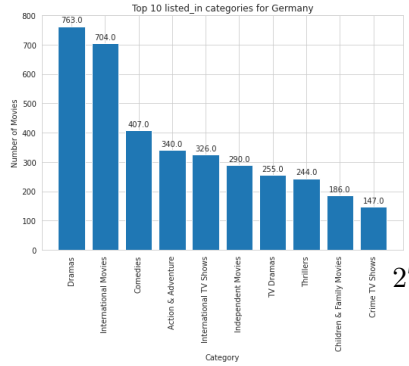

Top 10 listed_in categories for Japan
Number of Movies
Category
International TV Shows
Anime Series
Anime Features
Action & Adventure
International Movies
Kids' TV
Children & Family Movies
TV Dramas
Romantic TV Shows
Dramas
3431.0
2048.0
914.0
823.0
810.0
309.0
238.0
218.0
209.0
208.0

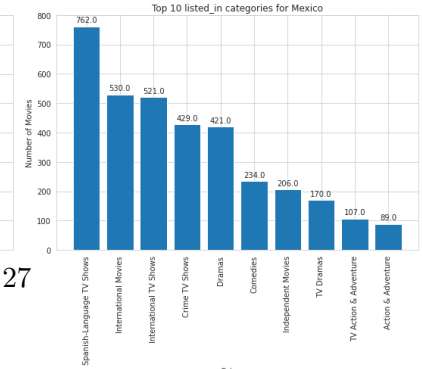

Top 10 listed_in categories for France
Number of Movies
Category
International Movies
Dramas
Independent Movies
Comedies
International TV Shows
Thrillers
Action & Adventure
Children & Family Movies
TV Dramas
Kids' TV
1740.0
1708.0
716.0
567.0
430.0
391.0
378.0
367.0
320.0
294.0


Top 10 listed_in categories for Canada
Number of Movies
Category
Comedies
Children & Family Movies
Dramas
Kids' TV
International Movies
Action & Adventure
Thrillers
Independent Movies
Horror Movies
TV Dramas
1017.0
919.0
793.0
534.0
437.0
432.0
415.0
392.0
363.0
343.0

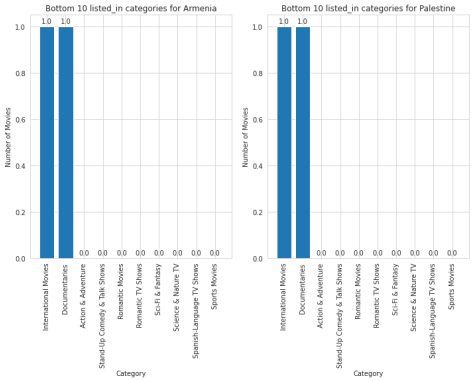

Top 10 listed_in categories for South Korea
Number of Movies
Category
International TV Shows
Romantic TV Shows
Korean TV Shows
International Movies
TV Dramas
Dramas
Comedies
TV Comedies
Crime TV Shows
Action & Adventure
1184.0
1094.0
1061.0
333.0
312.0
213.0
205.0
194.0
193.0
141.0


Top 10 listed_in categories for Spain
Number of Movies
Category
International Movies
Dramas
International TV Shows
Comedies
Spanish-Language TV Shows
Thrillers
Crime TV Shows
TV Dramas
Independent Movies
Romantic TV Shows
1131.0
691.0
597.0
476.0
467.0
319.0
284.0
173.0
153.0
120.0

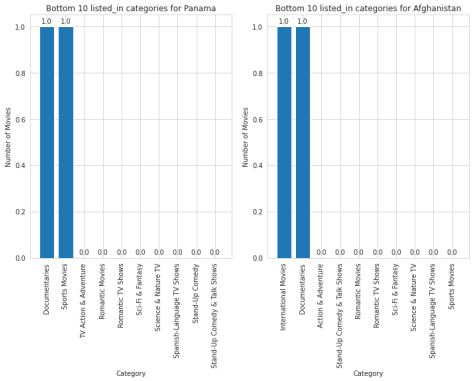

Top 10 listed_in categories for Germany
Number of Movies
Category
Dramas
International Movies
Comedies
Action & Adventure
International TV Shows
Independent Movies
TV Dramas
Thrillers
Children & Family Movies
Crime TV Shows
763.0
704.0
407.0
340.0
326.0
290.0
255.0
244.0
186.0
147.0


Top 10 listed_in categories for Mexico
Number of Movies
Category
Spanish-Language TV Shows
International Movies
International TV Shows
Crime TV Shows
Dramas
Comedies
Independent Movies
TV Dramas
TV Action & Adventure
Action & Adventure
762.0
530.0
521.0
429.0
421.0
234.0
206.0
170.0
107.0
89.0

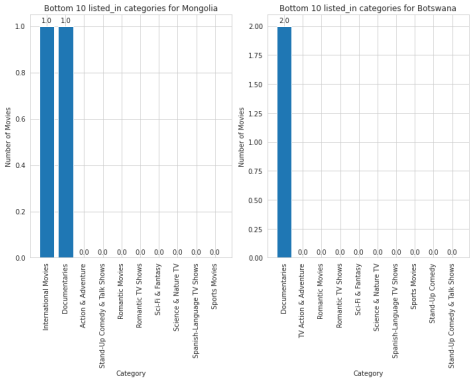
[562]:
```
import matplotlib.pyplot as plt

# Drop the 'Total_Movies' column from the DataFrame

fig, axes = plt.subplots(nrows=5, ncols=2, figsize=(10, 40))
for i, country in enumerate(bottom_10_countries.index):
    row = i // 2
    col = i % 2
    # select the top 10 listed_in categories for the current country
    top_10_categories = bottom_10_countries.loc[country].
 ↪sort_values(ascending=False)[:10]
    # create a bar chart for the current country
    bars = axes[row, col].bar(top_10_categories.index, top_10_categories.values)
    axes[row, col].set_title(f'Bottom 10 listed_in categories for {country}')
    axes[row, col].set_xlabel('Category')
    axes[row, col].set_ylabel('Number of Movies')
    axes[row, col].tick_params(axis='x', rotation=90)
    # add count labels to each bar
    for bar in bars:
        height = bar.get_height()
        axes[row, col].annotate(f'{height}', xy=(bar.get_x() + bar.get_width() /
 ↪ 2, height), xytext=(0, 3),
                                textcoords="offset points", ha='center',␣
 ↪va='bottom')
plt.tight_layout()
plt.show()
```


Bottom 10 listed_in categories for Armenia
Number of Movies
Category
International Movies
Documentaries
Action & Adventure
Stand-Up Comedy & Talk Shows
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies

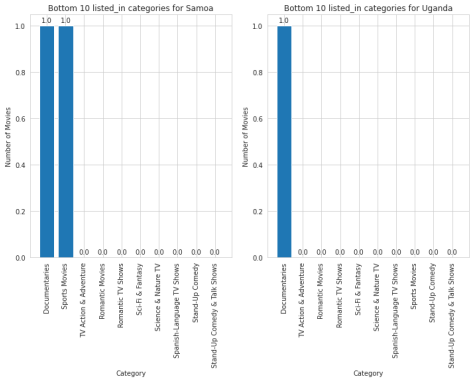
Bottom 10 listed_in categories for Palestine
Number of Movies
Category
International Movies
Documentaries
Action & Adventure
Stand-Up Comedy & Talk Shows
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies


Bottom 10 listed_in categories for Panama
Number of Movies
Category
Documentaries
Sports Movies
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Stand-Up Comedy
Stand-Up Comedy & Talk Shows

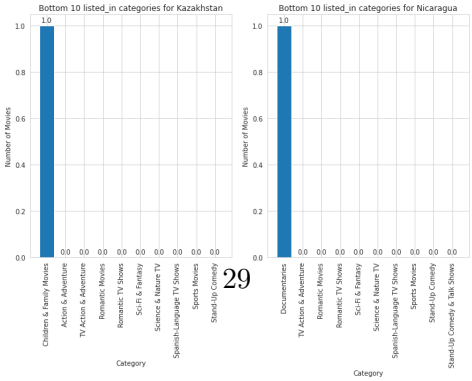
Bottom 10 listed_in categories for Afghanistan
Number of Movies
Category
International Movies
Documentaries
Action & Adventure
Stand-Up Comedy & Talk Shows
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies


Bottom 10 listed_in categories for Mongolia
Number of Movies
Category
International Movies
Documentaries
Action & Adventure
Stand-Up Comedy & Talk Shows
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies

Bottom 10 listed_in categories for Botswana
Number of Movies
Category
Documentaries
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies
Stand-Up Comedy
Stand-Up Comedy & Talk Shows


Bottom 10 listed_in categories for Samoa
Number of Movies
Category
Documentaries
Sports Movies
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Stand-Up Comedy
Stand-Up Comedy & Talk Shows

Bottom 10 listed_in categories for Uganda
Number of Movies
Category
Documentaries
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies
Stand-Up Comedy
Stand-Up Comedy & Talk Shows


Bottom 10 listed_in categories for Kazakhstan
Number of Movies
Category
Children & Family Movies
Action & Adventure
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies
Stand-Up Comedy

Bottom 10 listed_in categories for Nicaragua
Number of Movies
Category
Documentaries
TV Action & Adventure
Romantic Movies
Romantic TV Shows
Sci-Fi & Fantasy
Science & Nature TV
Spanish-Language TV Shows
Sports Movies
Stand-Up Comedy
Stand-Up Comedy & Talk Shows

[563]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 202061 entries, 0 to 202064
Data columns (total 15 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   show_id       202061 non-null  object
 1   type          202061 non-null  object
 2   title         202061 non-null  object
 3   director      202061 non-null  object
 4   cast          202056 non-null  object
 5   country       202061 non-null  object
 6   date_added    202061 non-null  datetime64[ns]
 7   release_year  202061 non-null  int64
 8   rating        202061 non-null  object
 9   duration      202061 non-null  object
 10  listed_in     202061 non-null  object
 11  description   202061 non-null  object
 12  day_of_week   202061 non-null  object
 13  year_added    202061 non-null  Int64
 14  month_added   202061 non-null  Int64
dtypes: Int64(2), datetime64[ns](1), int64(1), object(11)
memory usage: 33.1+ MB
```

[564]:
```
df['cast']
```

[564]:
```
0                    Adil Dehbi
1                    Ama Qamata
2                    Ama Qamata
3                    Ama Qamata
4                   Khosi Ngema
                   ...
202060            Anita Shabdish
202061            Anita Shabdish
202062    Chittaranjan Tripathy
202063    Chittaranjan Tripathy
202064    Chittaranjan Tripathy
Name: cast, Length: 202061, dtype: object
```

[565]:
```
famous_actor = df.groupby(['cast','country'])['title'].count().
 ↪reset_index(name='count').sort_values(by='count', ascending=False)
famous_actor[famous_actor.country == 'India'].sort_values(by='count',␣
 ↪ascending=False)
```

[565]:
```
                    cast country  count
4517         Anupam Kher   India    113
43079       Radhika Apte   India    101
48242     Shah Rukh Khan   India     99
38286  Naseeruddin Shah    India     96
1318        Akshay Kumar   India     87
…                    …      …     …
40957       Pascal Atuma   India      1
2114       Alhaji Fadika   India      1
2028     Alexei Maklakov   India      1
11839     Daniel Holguín   India      1
5680        Aída Morales   India      1

[5078 rows x 3 columns]
```

[566]:
```
indian_actors = df[df['country'] == 'India']
famous_actor = df.groupby(['cast','country'])['title'].count()
max_films_actor = famous_actor.idxmax()
# # Printing the results
print(max_films_actor)
```

```
('Belén Cuesta', 'United States')
```

6.1 Comments on the range of attributes

0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	category
6	date_added	8797 non-null	datetime64[ns]
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	category
11	Description	8807 non-null	object

Country and listed_in are added as category type because they are repeated and the multiple time the main reason of converting it from object into category type is Memory efficiency, Performance improvements, Improved functionality

The date_added is converted to time stamp in order to extract month day year which helps in analysis

show_id is a unique variable

Type deals with only 2 a)MOVIE b)TV SHOW

In this dataset we deal mostly with categorical data

6.2 Comments on the distribution of the variables and relationship between them

Box plots, Countplots, Bar plots are used in this project for univariate and bivariate analysis

sns.distplot(df['release_year']) generates a histogram with the frequency of movies released in each year on the y-axis and the release year on the x-axis. The KDE plot is a smoothed curve that represents the probability distribution of the release year. We can see the concentration of movies released in the more recent years, with a peak around 2018.

6.3 Comments for each univariate and bivariate plot

Univariate analysis:

From the box plot which represents the duration in minutes we can say the average duration of movies is around 106 minutes and max duration of a movie is 312 minutes and min duration is 3 minutes

Highest Length Movie :- Black Mirror: Bandersnatch

Lowest Length Movie :- Silent

The best time to release a TV show is Friday

The frequency of Movies and TV shows are represented through dist plot The Movies growth rate is always good. But the Growth rate has been improved in the 21st century for TV shows when compared to movies.

These are the Top 10 content generating countries

'United States', 'India', 'United Kingdom', 'Japan', 'France', 'Canada', 'South Korea', 'Spain', 'Germany', 'Mexico'

Bivariate analysis:

I used Countplot to Check the growth rate of Movies and TV shows per release year
Its graph tells us the Movies Growth rate had a dip in 2018 and TV shows growth rate has been steadily increasing

Pie charts were used to understand the directors who were a part of only movies,TV shows and both

Pie charts were used to understand the cast who were a part of only movies,TV shows and both

Business Insights

- **INDIA's CONTENT IS MOSTLY FOCUSSED ON TV-PG: TV Parental Guidance Suggested ,TV-MA: TV Mature Audience,TV-14: TV 14 and older**
- **FOR General Audiences USA made more movies followed by UK and Spain**
- **International Movies has more demand in INDIA followed by France and UK**
- **Netflix has 72% content filled with movies rest deals with TV Shows**
- **Uruguay Country produces very less TV SHOWS Kazakhstan,Nicaragua,Uganda stands in the bottom place with the same record of producing movies**
- **USA,INDIA,UK are in Top3 in producing Movies and United-States,Japan,UnitedKingdom,South,Korea,Mexico are in top 5 in producing TV – SHOWS**
- **The Highest Movies Acted Cast is From United States and his name is Belén Cuesta has done nearly 167**
- **The Lowest Movies Acted Cast is From United States and his name is Belén Cuesta has done nearly 167**
- **Anupam Kher,Radhika Apte,Shah Rukh Khan are most famous actors in INDIA they acted in more than 90 films**

Recommendations:

- **More number of TV Shows should be added in the geographies like South Korea, Japan, UK.**
- **Most viewership of TV Shows are till 4 -5 seasons only, in almost every genre. So adding more number of seasons in each category won't provide positive result**
- **Developing markets like India, Japan, and Canada are content consuming both in Movies and in TV Shows market with descent enough duration. Loading more/ trying new content will make sense.r**
- **There are few countries where there is no response in terms**

- **International Movies have high demand in almost every countries Netflix should promote more international films where its response is very less like Panama & Botswana**
- **As the Viewership is very less in those countries it should try to deliver good and famous content to its viewers slowly**
- **Netflix should try to make people engage with its platform for example it should try to make more documentaries where people know the real truth.**
- **Countries like USA INDIA South Korea are more engaging if the content is good it should try to deliver more sequals or seasons to make the audience interactive.**